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THE DISTANCE IN **LIGHT-YEARS**
TO THE **EAGLE NEBULA****IN APRIL 1995, NASA**

AMALGAMATED 32 individual Hubble Space Telescope photographs to produce an image known as **The Pillars of Creation**. It shows immense clouds of interstellar gas and dust several light-years long, shaped by intense ultraviolet radiation from nearby stars, in which **new stars are forming**.

Also in space, NASA's Galileo spacecraft dropped a probe into Jupiter's atmosphere. Analysis of **meteorite ALH84001**, which originated from Mars and was discovered in Antarctica in 1984, revealed tiny structures that resembled fossilized bacteria. The discovery prompted speculation that these were the first **definitive signs of extraterrestrial life**. However, further analysis has since shown that the structures are almost certainly not of biological origin.

**Wireless endoscopy**

The **PillCam** is a swallowable capsule that contains a tiny camera, a flashing light, and a radio transmitter to send images to a receiver.

Pillars of Creation

In this dramatic composite image, areas of star birth in the Eagle Nebula can be seen in the tiny globules at the tops of the pillars.

In October 1995, astronomers detected a planet orbiting the star 51 Pegasi, by measuring the wobble in the star's motion caused by the planet's presence. It was the **first confirmed planet around an ordinary star** other than the Sun.

Israeli inventor **Gavriel Iddan** patented the **PillCam** in 1995. It is a small capsule that a patient swallows, and which then passes through the digestive tract taking pictures and sending them wirelessly to a receiver. The PillCam gave gastroenterologists a new, safe, low-cost window on processes and problems in the large section of the digestive system **where endoscopes cannot reach**.

There were three important advances in modern physics in 1995. At the time, theoretical physicists had developed five **separate superstring theories**, which propose that the particles of matter and force are actually **tiny vibrating one-dimensional objects** (strings). Each theory assumed the existence of several dimensions in addition to the three space dimensions with which we are familiar in everyday life. The extra dimensions are



January 17, 1995 Patent filed for wireless capsule endoscopy, involving a swallowable "pill camera"

April 1, 1995 Hubble Space Telescope reveals star formation in progress in the Eagle Nebula

June 5, 1995 Creation of first Bose-Einstein Condensate, a new state of matter

May 1995 Ed Witten develops M-theory, by uniting superstring theory and supersymmetry

July 13, 1995 A NASA space probe descends into Jupiter's atmosphere

July 28, 1995 A team at The Institute for Genomic Research announce the first sequencing of a bacterial genome

October 6, 1995 First confirmed detection of a planet orbiting an ordinary star other than the Sun

September 1995 CERN produces the first antiatoms: atoms consisting of antiprotons and antielectrons (positrons)

December 8, 1995 NASA's Galileo space probe becomes the first spacecraft to orbit Jupiter

December 1995 The Hubble Space Telescope captures its celebrated Hubble Deep Field image